

Question paper contains two parts.
Answer all questions.
Answer all parts of a question together.

PART A (Closed Book)

1. Select the most appropriate alternative:

i) Pascal second is unit of

- (a) pressure
- (b) kinematic viscosity
- (c) dynamic viscosity
- (d) surface tension

ii). Viscosity of gas

- (a) decreases with increase in temperature
- (b) increases with increase in temperature
- (c) is independent of temperature
- (d) none of the above

iii). The intensity of pressure developed by surface tension of 0.075 N/m in a droplet of water of 0.075 mm diameter is

- (a) 0.8 N/cm²
- (b) 0.6 N/cm²
- (c) 0.4 N/cm²
- (d) 0.5 N/cm²

iv). A vertical triangular area with vertex downward and altitude "h" has its base lying on the free surface of liquid. The centre of pressure below the free surface is at a distance of

- (a) h/6
- (b) h/3
- (c) h/2
- (d) 2h/3

v). Centre of buoyancy always

- (a) coincides with the centre of gravity
- (b) coincides with the centroid of the volume of fluid displaced
- (c) remains above the centre of gravity
- (d) remains below the centre of gravity

vi). When a liquid rotates at a constant angular velocity about a vertical axis as a rigid body, the pressure intensity varies

- (a) linearly with radial distance
- (b) as the square of radial distance
- (c) inversely as the square of radial distance
- (d) inversely as the radial distance

vii). The equation $P/\gamma + v^2/2g + Z = \text{constant}$ is based on the following assumptions regarding the flow of liquid

- (a) steady, frictionless, incompressible and along a streamline
- (b) steady, frictionless, uniform and along a streamline
- (c) steady, incompressible, uniform and along a streamline
- (d) steady, frictionless, uniform and incompressible

viii). The continuity equation $\rho_1 V_1 A_1 = \rho_2 V_2 A_2$ is based on the following assumptions regarding the flow of liquid

- (a) steady flow
- (b) uniform flow
- (c) incompressible flow
- (d) frictionless flow

ix). Which of the following velocity potentials satisfies continuity equation

- (a) $x^2 y$
- (b) $x^2 - y^2$
- (c) $\cos x$
- (d) $x^2 + y^2$

x). The magnitude of component of velocity at point (1,1) for a stream function $\psi = x^2 - y^2$ is equal to

- (a) 2
- (b) $2\sqrt{2}$
- (c) 4
- (d) 5

xi). Equation of continuity is based on the principle of conservation of

- (a) mass
- (b) energy
- (c) momentum
- (d) none of the above

xii). A fluid jet discharging from a 4 cm diameter orifice has a diameter 3 cm at its vena contracta. If the coefficient of velocity is 0.98, the coefficient of discharge for the orifice will be

- (a) $0.98 \times (0.75)^2$
- (b) $(0.75)^2 / 0.98$

(c) $0.98 \times (1)$
(d) $0.98 / (1)$

xiii). Due to

- (a) 5 %
- (b) 10 %
- (c) 15 %
- (d) 20 %

xiv). The dis

- (a) $H^{1/2}$
- (b) $H^{3/2}$
- (c) $H^{7/2}$
- (d) none of the

xv). $\Delta\psi$ betw

- (a) velocity
- (b) discharge
- (c) head
- (d) pressure

xvi). A dime

- (a) $\sigma D / \rho V$
- (b) $\sigma D / \rho V^2$
- (c) $\sigma / D \rho V^2$
- (d) none of the

xvii). An erro

- (a) 0.5 % in t
- (b) 1.0 % in t
- (c) 1.25 % in
- (d) 1.5 % in t

xviii). Pressur

where x is eq

- (a) 11.62 m
- (b) 11.92 m
- (c) 12.82 m
- (d) 13.12 m

xix). A rectan

0.5 m, if wat

- (a) 5 kN
- (b) 10 kN
- (c) 15 kN

- (c) $0.98 \times (1.33)^2$
(d) $0.98 / (1.33)^2$

xiii). Due to each contraction, the discharge of rectangular sharp crested weir is reduced by

- (a) 5 %
(b) 10 %
(c) 15 %
(d) 20 %

xiv). The discharge over a V-notch varies as

- (a) $H^{1/2}$
(b) $H^{3/2}$
(c) $H^{7/2}$
(d) none of the above

xv). $\Delta\psi$ between two stream lines represents

- (a) velocity
(b) discharge
(c) head
(d) pressure

xvi). A dimensionless combination of surface tension σ , density ρ , diameter D and velocity V is

- (a) $\sigma D / \rho V$
(b) $\sigma D / \rho V^2$
(c) $\sigma / D \rho V^2$
(d) none of the above

xvii). An error of 0.5% in the measurement of head in a V-notch cause an error of

- (a) 0.5 % in the discharge
(b) 1.0 % in the discharge
(c) 1.25 % in the discharge tank
(d) 1.5 % in the discharge

xviii). Pressure of 20 kPa is equivalent to a head of x meters of carbon tetra-chloride of relative density 1.59 where x is equal to

- (a) 11.62 m
(b) 11.92 m
(c) 12.82 m
(d) 13.12 m

xix). A rectangular block 2 m long, 1 m wide and 1 m deep floats in water, the depth of immersion being 0.5 m, if water weighs 10 kN/m^3 , then the weight of block is

- (a) 5 kN
(b) 10 kN
(c) 15 kN

(d) 20 kN

xx). Size of venturimeter is specified by

- (a) pipe diameter
- (b) throat diameter
- (c) angle of diverging section
- (d) both pipe diameter as well as throat diameter

2. (a) What do you understand by Cipolletti weir and derive an equation to measure the discharge over it.
(b) Define Broad Crested Weir and find the critical depth for which discharge over is maximum.
(c) Derive Euler's equation.
3. (a) The resistance force F of a ship is a function of its length L , velocity V , acceleration due to gravity g and fluid properties like density ρ and viscosity μ . Write this relationship in dimensionless form. (Use Buckingham Pi theorem, take L , V and ρ as repeating variables).
(b) A 90° conical vessel of height 2 m carries water whose volume is one half of the volume of the cone. Find the speed at which the vessel may be rotated to make the water just spill out.
(c) The velocity distribution for laminar flow between parallel plates is given by

$$\frac{u}{u_{\max}} = 1 - \left(\frac{2y}{h}\right)^2$$

Where h is the distance separating the plates and the origin is placed midway between the plates. Consider a flow of water at 15°C , with $u_{\max} = 0.10\text{ m/s}$ and $h = 0.1\text{ mm}$. Calculate the shear on the upper plate and give its direction. Sketch the variation of shear stress across the channel. ($\mu = 1.14 \times 10^{-3}\text{ NS/m}^2$)

4. A turbine has a supply line of diameter 40cm and a tapering draft tube as shown in fig. When the flow in the pipe is $0.5\text{ m}^3/\text{s}$ the pressure head at the point A upstream of turbine is 30 m and at point B in the draft tube, where the diameter is 60cm, the pressure head is (-4m). point B is 2.0 m below point A. Estimate the power output of the turbine by assuming 90% efficiency. (10)

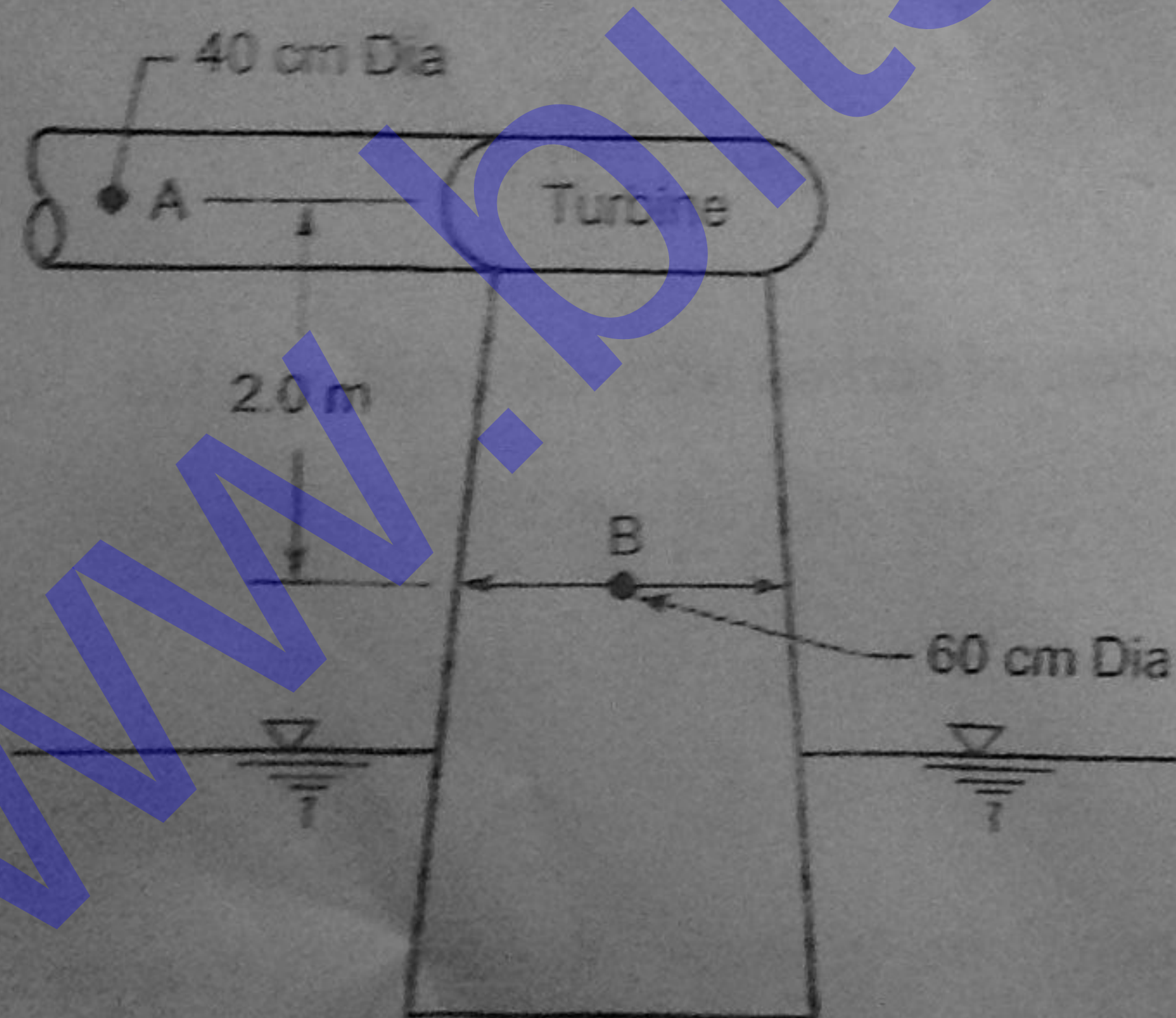


Figure Q4